

The first objection is based on certain interpretations of **quantum mechanics**, our best current theory of the physical world.

According to the standard interpretations of quantum mechanics – though not all interpretations of the theory – the physical world is indeterministic.

That is, what the laws of nature tell us in many cases is not what will happen, but rather just probabilities of various outcomes.

Imagine, for example, that we have some uranium, and we are interested in whether some particular atom will decay in a certain amount of time.

Suppose now that the atom does decay, and we ask: **why** did it decay?

Quantum mechanics will tell us something like this: there is a 12.37% chance that the atom will decay in that amount of time.

On standard interpretations of quantum mechanics, there is literally no answer to this question. There was a chance that it would, and a chance that it wouldn't, and it just did — end of story. Is that a problem for the PSR?

It appears so. For it looks like this would imply that there is a contingent fact -- namely, that the atom would decay at that moment -- which has no explanation.

The defender of the cosmological argument can always reply that this shows that quantum mechanics is incomplete. And this is not an entirely unreasonable thing to say. Quantum mechanics is inconsistent with our other best theory of the physical world -- the general theory of relativity -- which suggests that the correct complete theory of the physical world may not look quite like any of our current theories.

But in response to this reply to the objection, there are two things
to be said.

The first is simple. The fact that a premise of an argument conflicts with well-supported claims of our best scientific theory is a strike against that premise. Whether or not it convinces us to reject the premise, it seems that it should at least reduce our confidence in it.

The second reply to the reply is more abstract.

First, notice that the current objection to the PSR does not depend on any details of quantum mechanics. It just depends on the idea that our best physical theory is indeterministic.

So it seems that if you defend the PSR, you must think that the correct theory of the physical world is a deterministic theory -- one on which the laws of nature, plus the state of the world, determine what will happen in the future.

Here's the problem. As we'll see in the next section of the course, a plausible argument can be made that determinism is inconsistent with the existence of free will.

But free will is a big part of the most popular response to the main argument for atheism -- the argument from evil.

So it looks like the defender of the cosmological argument might need to think hard about whether her picture of the world makes room for freedom of the will, and evil.



Leibniz's
argument



the
PSR



objections
to the
PSR

The first objection is based on certain interpretations of **quantum**

mechanics, current theory of the physical world.

According to standard interpretations of quantum mechanics - though

not all interpretations of the theory - the physical world is indeterminate

happen, but rather just probabilities of various outcomes.

That is, what the laws of nature tell us in many cases is not what will

Imagine, for example, that we have uranium, and

we are interested in whether some particular will

decreasing in a certain amount of time.

is a 12.37% chance that the atom will decay in that

announcements.

Quantum mechanics will tell us something like this: there

On standard interpretations of quantum mechanics, there is literally

no answer to this question. There was a chance that it would, and a

problem for the PSR?

chance that it wouldn't, and it just did—end of story. Is that a

It appears so. For it looks like this would imply that there is a

contingent fact -- namely, that the atom would decay at that

moment--which has no explanation.

The defender of the cosmological argument always reply that

an entirely unreasonable thing to say. Quantum mechanics is

this shows that quantum mechanics is incomplete. And this is not

general theory of relativity -- which suggests that the correct

inconsistent with our best theory of the physical world - the

complete theory of the physical world may not look quite like any of

our current theories.

to be said.

But in response to this reply to the objection, there are two things

with well-supported claims of our best scientific theory is a strike

premise, it seems that it should at least reduce our confidence in it.

The first is simple. The fact that a premise of an argument conflicts

against that premise. Whether or not it convinces us to reject the

The second reply to the reply is more abstract.

First, notice that the current objection to the PSR does not

depend on any details of quantum mechanics. It just depends on

the idea that our best physical theory is deterministic.

So it seems that if you defend the PSR, you must think that the

on which the laws of nature, plus the state of the world,

determine what will happen in the future.

correct theory of the physical world is deterministic theory --

courts, a plausible argument can be made that determinism is

Here's the problem. As we'll see in the next section of the

inconsistent with the existing free will.

main argument for atheism -- the argument from evil.

But free will is a big part of the most popular response to the

room for freedom of the will, and evil.

So it's like the defender of the constitutional argument might

need to think hard about whether her picture of the world makes

Last time we considered an objection to Aquinas' assumption that if a first

be Good.

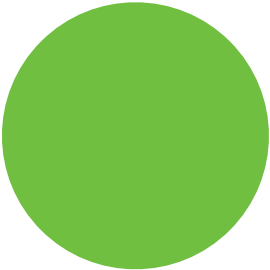
simply be some event, like the Big Bang, which is not a plausible candidate to

cause exists, then God exists: the objection was that the first cause could



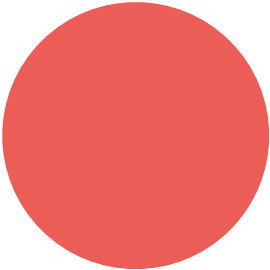
Leibniz's
argument







Leibniz's
argument







the
PSR



objections
to the
PSR

... we cannot find in any of the individual things, or even in the entire collection and series of things, a sufficient reason for why they exist.

Let us suppose that a book on the elements of geometry has always existed, one copy always made from another. It is obvious that although we can explain a present copy of the book from the previous book from which it was copied, this will never lead us to a complete explanation, no matter how many books back we go, since we can always wonder why there have been such books.

What is true of these books is also true of the different states of the world ... however far back we might go into previous states, we will never find in those states a complete explanation for why there is a world at all, and why it is the way it is.



